

# Creating a simplified prediction market with ECR20

Gambling ääähm “investing” has never been easier!

Wallet

• Hardhat localhost

Reconnect Wallet

|              |                |
|--------------|----------------|
| Address      | 0xF407...2a473 |
| Chain        | 0xaa36a7       |
| VOTE balance | 1000           |
| Allowance    | 10             |

Claim 1000 VOTE once per wallet

# Storyline

## 1 Prediction markets

Why pooling stakes can look like collective belief.

## 2 Our DApp

VOTE token, YES/NO pools, MetaMask, Sepolia.

## 3 Live demo

A funny race market: predict a marble winner.

## 4 Security risks

Admin oracle, ambiguity, incentives, and demo limits.

# What is a prediction market?

## EXAMPLE QUESTION

**Will LEC be the first marble to reach the goal?**

YES if LEC crosses the finish line first.  
NO if any other marble finishes first.

## Pool split

Probability display in our MVP is just the stake distribution, not a truth guarantee.

YES

68%



680 VOTE

NO

32%



320 VOTE

# Known providers



Polymarket

**Kalshi**

# Some funny bets

“How many times Elon Musk will tweet within a week”

“Is the Earth flat?”

“Will Jesus Christ return before GTA VI?”

“Will the US confirm that aliens exist in 2026?”

“Will Trump do a podcast with Hawk Tuah Girl?”

“Will the HBO documentary identify Elon Musk as Satoshi Nakamoto?”

“Will AI be the 2024 TIME Person of the Year?”

# One might be thinking



# Current affairs

test ▾ Local News ▾ • Live ▾ Shows ▾ ... CBS NEWS

Minutes Overtime | Overtime Latest Original Rewind 60 Minutes

60 Minutes Overtime

## Suspected insider accounts net \$2.4 million on Polymarket Iran war bets with 98% win rate, firm finds

By [Jon Wertheim](#), [Aliza Chasan](#), Andrew Bast, Jessica Kegu

May 17, 2026 / 7:17 PM EDT / CBS News

[Add CBS News on Google](#)

Nine connected Polymarket accounts have raked in more than \$2.4 million betting almost exclusively on U.S. military actions in what

## Zürcher Google-Angestellter wegen Insiderwetten auf Polymarket verhaftet

Welche Person wurde 2025 am meisten gegoogelt? Dieser Ingenieur wusste das vor der Öffentlichkeit – und machte die Information auf einer Wettplattform zu Geld. Es ist die zweite Festnahme wegen Insidertrading auf Prognosemärkten.

Ruth Fulterer 29.05.2026, 16.35 Uhr 4 Leseminuten

Bets

### PRESS RELEASE

## U.S. Soldier Charged With Using Classified Information To Profit From Prediction Market Bets

Thursday, April 23, 2026

Share >

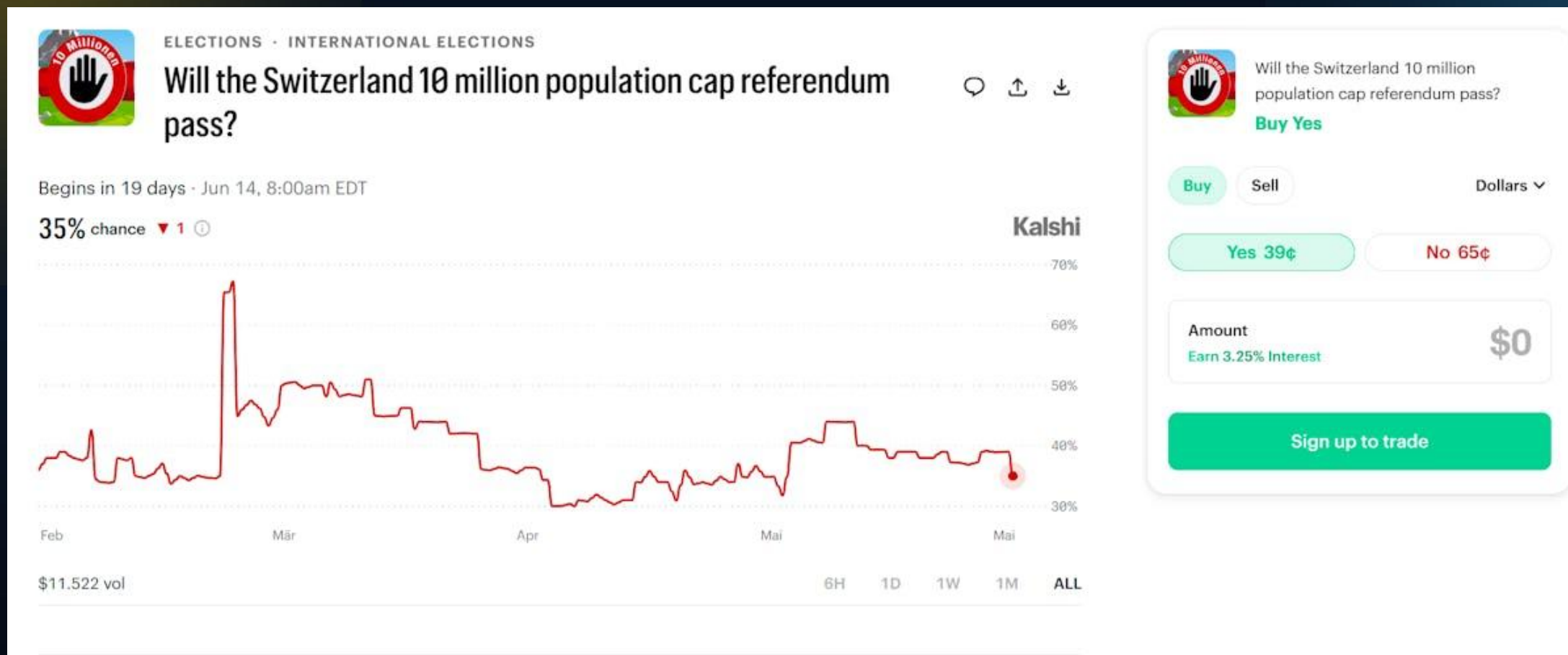
For Immediate Release

Office of Public Affairs

Gannon Ken Van Dyke allegedly made more than \$400,000 trading on polymarket on the basis of classified information regarding the timing of a U.S. military operation to capture Nicolás Maduro in Venezuela

The Justice Department announced today the unsealing of an indictment charging Gannon Ken Van Dyke, a U.S. Army soldier, with unlawful use of confidential government information for

# In Switzerland





# In Switzerland

Politik · Referendum

## Juni-Referendum in der Schweiz: Was wird passieren?

Vergangen ▾ **Juni 14**

\$135,510 Vol. | 14. Juni 2026

Polymarket

|                                                                                                                            |                  |                      |                       |
|----------------------------------------------------------------------------------------------------------------------------|------------------|----------------------|-----------------------|
| <b>Zivildienstgesetz</b><br>\$8,054 Vol.  | <b>74%</b> ▲ 21% | <b>Kaufen Ja 76¢</b> | <b>Kaufen ... 28¢</b> |
| <b>Nein zu zehn Millionen Schweiz</b><br>\$127,768 Vol.                                                                    | <b>39%</b> ▲ 4%  | <b>Kaufen Ja 39¢</b> | <b>Kaufen ... 62¢</b> |

**Regeln** Marktkontext

As of market creation, two referenda are scheduled to be held in Switzerland on June 14, 2026 :

- Issue 1: No to ten million Switzerland
- Issue 2: Referendum on the Civilian Service Act

This market will resolve to "Yes" if the listed referendum is approved under the applicable legal threshold in the nationwide popular vote held in the specified popular vote. Otherwise, this market will resolve to "No."... [Mehr anzeigen](#)

# Current affairs

The ISW insisted it did “not engage with the users or representatives of such platforms”.

“We strongly condemn and have never consented to the exploitation of our products for the abhorrent purpose of gambling on war,” said a spokesperson.

‘No one wants to make money off war or people dying, but ...’

“I love to gamble,” said Joseph Francia.

Now in his early 30s, Francia counted cards in casinos while studying economics at Berkeley, and spent weekends in Reno, Nevada, playing blackjack. He’s not a thrill-seeking “Yolo” (you only live once) gambler, he said: he likes to bet when he has an edge on the house.

At university, he and a friend decided to collect data from a number of offshore sportsbooks, and start placing arbitrage bets: playing on the discrepancies in odds given by different betting sites.

# Buuuuuuut



# Our DApp in one screen

SIMPLIFIED PREDICTION MARKET

## Shared VOTE, pooled outcomes, and admin resolution.

Claim VOTE from the faucet, create a market, stake on YES or NO, and let the admin resolve after the deadline. Probabilities come directly from the pool split.

● Wallet + Faucet

Wallet

Hardhat localhost



Reconnect Wallet

Address

0x0f407fa744f07ecf022b6052625f84c08412a473

Chain

0xaa36a7

VOTE balance

0

Allowance

0

Claim 1000 VOTE

● Admin resolution

● Pool split

MARKETS

All live and resolved markets

Admin: 0x0F407FA744f07eCf022b6052625F84C08412a473

Market #0

will it be 32degree on 28.05 in bern?

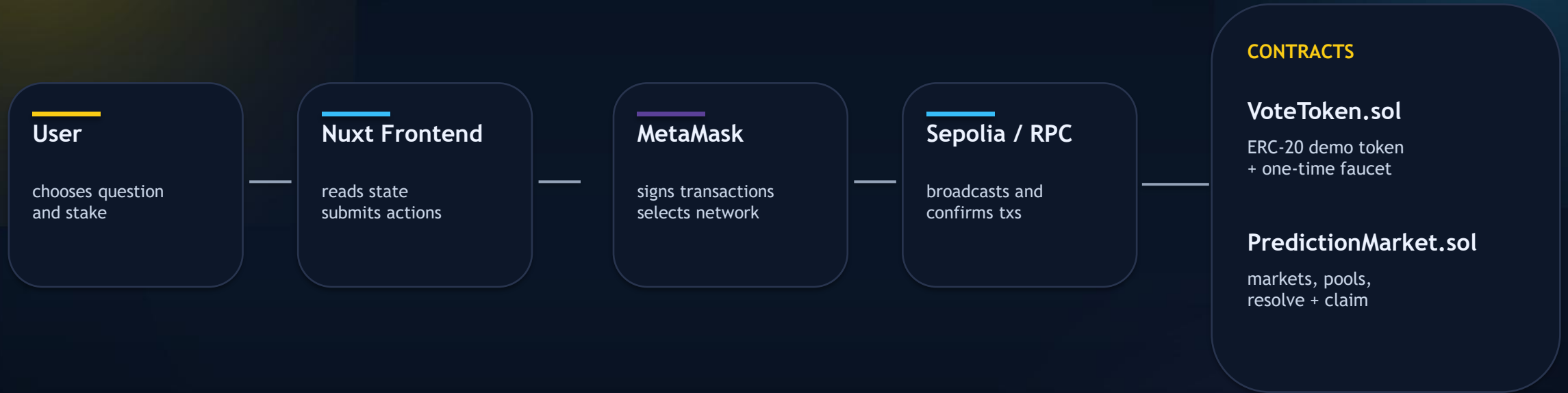
Resolved

# User flow



Each state change is a signed transaction. The frontend is only the interface.

# Architecture



**Important: the blockchain is the source of truth for balances, pools and resolution state.**

# Smart contract mechanics

## VoteToken.sol

- ERC-20 token named VOTE
- Faucet mints 1000 VOTE
- One claim per address
- No real financial value / no stable coin
- Aprove(Marked, 100)

## PredictionMarket.sol

- createMarket(question, context)
- stake(YES, markedId, 100)
- resolveMarket(winner)
- claimReward()

## Payout formula

$$\text{reward} = \text{userWinningStake} \times \text{totalPool} / \text{winningPool}$$

The losing side funds the winning side proportionally.

• YES / NO fixed in the UI for the MVP

• Contract supports 2+ outcomes

• Admin-only resolution

• Pull-pattern claim

# Live demo: marble race market

## MARKET QUESTION

Will LEC be the first marble to finish?



## Resolution rule

Only the selected marble crossing the finish line first counts.  
If unclear, replay frame-by-frame before admin resolves.



# Demo run sheet

**1** Open the deployed DApp  
Show wallet + market list

**2** Claim / confirm VOTE balance  
Use prepared wallets

**3** Create or open the market  
Question + strict context

**4** Audience predicts  
YES / NO by hand vote

**5** Stake VOTE  
Approve → Stake

**6** Play the race  
Identify winner together

**7** Admin resolves  
YES or NO after deadline

**8** Winner claims reward  
Show pull payout

Backup plan: if the network is slow, show an already resolved market and still discuss oracle/admin risk.

# Security Risks I: The truth Layer

## Centralized Admin Oracle

Only the admin can resolve a market. A wrong or malicious result is technically final in this MVP.

## Ambiguous Rules

A race needs exact rules: selected marble, finish line, camera angle, ties, and replay frame.

## Insider Information

Anyone who has watched the race already has a strong advantage over honest guessers.

**Mitigation for the demo: strict resolution context, public replay, and clear disclosure that admin resolution is centralized.**

# Security risks II: Implementation & UX

## Faucet Sybil

One claim per address is easy to bypass by creating more wallets. Fine for a demo, weak for real incentives.

## Allowance flow

approve + transferFrom is normal ERC-20 UX, but users can accidentally approve the wrong amount or contract.

## Public stakes & Staker list scaling

All stakes are visible. Late participants can react to the pool and strategically flip outcomes.  
Tracking all stakers and returning arrays is convenient, but can become expensive or impractical at scale.

## Front-Running

FTransactions are visible before they're processed

● hasClaimed[address]

● transferFrom()

● userStakes public

● getMarketStakers()

# Security posture of the MVP

## Good MVP choices

- Pull-pattern claim avoids bulk payouts
- claimed flag set before token transfer
- No real-money collateral
- Short, inspectable smart contracts
- Transparent on-chain pool state

## Still demo-only

- Central admin decides truth
- No Chainlink / oracle integration
- No dispute or appeal period
- No market moderation
- No secondary trading or liquidity model

# Future work

## Decentralized resolution

Oracle, reporter jury or challenge mechanism.

## Dispute period

Delay payout until objections are settled.

## Market moderation

Prevent unclear or maliciously framed markets.

## Privacy / commit-reveal

Hide votes until after market close for sensitive markets.

## Scalable indexing

Use events + off-chain indexer instead of reading large arrays.

## Gas analysis

Measure approve, stake, resolve and claim costs.

Most improvements are not about making the UI prettier; they are about making “truth” less centralized.

## TAKEAWAY

**The smart contract can  
count tokens.  
It cannot see which  
marble crossed first.**

### Demo message

1. Users can stake.
2. Pools are transparent.
3. Admin resolves.
4. Oracle problem remains.

● Questions?

Prediction markets are technically simple to demonstrate — but socially and economically hard to make trustworthy.